

Executive Performance & Revenue Intelligence Report

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From: Business Intelligence & Revenue Operations Team

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Subject: NovaTech Unified Revenue Architecture, Operational Insights, and Generative BI

Executive Summary

NovaTech's revenue operations were reviewed across three main areas: marketing lead acquisition, sales pipeline conversion, and customer support. Across 2,240 leads, \$12.36M in marketing spend yielded \$1.13M in directly attributed revenue. The sales pipeline processed 499 closed deals, generating \$0.71M in closed won revenue at a 63.13% win rate with an average sales velocity of 67 days. On the customer health side, 3,000 support tickets averaged 58.2 hours to resolve, with enterprise accounts logging 757 tickets characterized by a 25.5% negative sentiment rate and critical product incidents reaching resolution times over 102 hours.

This report explains the data consolidation strategy, the three-sheet QuickSight dashboard, the impact of Generative BI (QuickSight Q), and recommendations to reduce operational leakage and protect annual recurring revenue (ARR).

1. Data Strategy & Architecture

NovaTech previously operated across isolated source systems: marketing campaign trackers, CRM pipeline stages, and customer service desks. This fragmentation made it difficult to connect acquisition channels with long-term customer value.

- **Anchor Schema Design:** The data architecture established `novatech_crm_deals` as the transactional backbone. Using left outer joins, `novatech_marketing_campaigns` and `novatech_support_tickets` were unified on the primary entity key `account_id`.
- **Grain Integrity & SPICE Ingestion:** Joining individual ticket events (3,000 records) to CRM deals (499 records) created a risk that deal amounts could be counted more than once during aggregation. By staging calculations in SPICE and maintaining dedicated single-domain views alongside the NovaTech Unified Dataset, we kept financial and operational metrics accurate.
- **Calculated Business Metrics:** Core key performance indicators—such as marketing return on investment, sales velocity (`days_to_close`), binary win conversion (`is_won_deal`), and ticket resolution durations—were standardized directly inside the data model to keep formulas consistent across BI visuals and natural-language queries.

2. Dashboard Design Rationale

The executive dashboard has three functional views that guide leadership from acquisition to customer retention:

- **Sheet 1: Marketing Funnel** — Purpose: Evaluate acquisition efficiency and spend-to-revenue ratios across campaigns and channels. Design: High-level KPI summary cards anchor gross spend (\$12.36M) and attributed returns (\$1.13M). Funnel progression and clustered bar visuals highlight lead drop-off and channel response divergence. Interactive controls allow immediate filtering by campaign channel and customer segment.
- **Sheet 2: Sales Pipeline** — Purpose: Assess sales velocity, conversion probabilities, and primary pipeline loss factors. Design: Tracks realized deal value (\$0.71M) and conversion rates (63.13%). One-click visual actions enable executives to click the "Won" deal stage or a specific industry bar, such as Technology or Healthcare, to cross-filter the entire sheet in real time.
- **Sheet 3: Customer Health** — Purpose: Identify post-sale account friction, technical debt bottlenecks, and renewal vulnerabilities. Design: Displays SLA performance across product categories and sentiment distributions across subscription tiers. A cross-dataset visual links CRM contract values with support ticket volumes. A cross-sheet navigation action allows reviewers to click an at-risk enterprise account and jump directly to its pipeline history on the Sales sheet.

3. Generative BI: QuickSight Q Topic Configuration & Accuracy

To provide conversational self-service analytics, Amazon QuickSight Q was tested before and after Topic optimization.

- **Baseline Performance (Out-of-the-Box Q):** Queries regarding “win rate” failed or generated raw deal counts because the system looked for an exact column match rather than averaging the binary win indicator. Revenue inquiries prompted ambiguous disambiguation between marketing attributed revenue and closed deal values. Technical database keys such as account_id, opportunity_id, and ticket_id added noise to responses.
- **Topic Optimization Strategy:** The NovaTech Business & Revenue Intelligence Topic was engineered with four adjustments: excluding noisy keys; enriching core measures with business synonyms; mapping Win Rate directly to the average of is_won_deal formatted as a percentage; and adding natural-language rules that prioritize closed won revenue for sales questions and attributed returns for marketing queries.
- **Measured Impact on Accuracy:** Re-testing showed clear improvements. Prompts such as “Show me win rate by industry” returned an accurate visual ranking matching the 63.13% benchmark. Revenue-channel queries grouped revenue_attributed without manual disambiguation, and comparative campaign filters were correctly computed.

4. Generative AI vs. Dashboard Comparative Analysis

To test QuickSight Q for executive self-service, its natural-language answers were compared with the dashboard results.

- **Where AI agreed with the Dashboard:** QuickSight Q achieved full alignment on single-domain aggregations, rankings, and dimensional segmentations. Direct Mail was correctly identified as the highest converting channel (53.0%), while Email lagged at 8.7%. Enterprise accounts yielded the highest average deal value. Customer-support duration splits were also accurately parsed, with critical tickets averaging 61.3 hours versus 58.4 hours for low-priority tickets.
- **Where AI disagreed or struggled:** QuickSight Q had difficulty with multi-metric queries across different data levels. It could not list the top 10 accounts by support ticket volume alongside their total CRM deal revenue.
- **Grain and Visual Scope Limitations:** The Customer Health sheet pre-aggregates operational data at higher categorical grains rather than the individual account level, leaving no account-grain visual for direct cross-checking.
- **Cross-Entity Multi-Metric Joins:** Combining CRM deal revenue with support-ticket counts at account level requires complex joins across tables and careful handling of data levels. Without a pre-built account summary table, Q could not resolve the dual-metric ranking.
- **When to use Q vs. the published dashboard:** Use QuickSight Q / Quick Chat for ad-hoc comparisons, flexible filtering, and self-service analysis by non-technical users. Use the published dashboard for regular executive monitoring, decision-making, and guided cross-sheet workflows.

5. Key Findings & Strategic Recommendations

Area	Quantified Finding	Business Impact	Strategic Action
Marketing Efficiency	\$12.36M spend yielded \$1.13M revenue. Direct Mail converts at 53.0% vs. Email at 8.7%.	Low-touch email blasts dilute budget; high-touch targeted channels convert.	Reallocate \$500K from email to direct mail.
Sales Conversion	63.13% overall win rate. Pricing & Feature Gaps account for >60% of 184 lost deals.	Enterprise deals stall in late stages despite delivering highest ACV.	Deploy deal-desk pricing waivers & battle cards.
Customer Retention	3,000 tickets averaged 58.2h MTTR; Analytics Dashboard criticals exceeded 102h SLA.	Product instability in key areas threatens enterprise ARR renewals.	Build Tier-3 escalation pod; cap MTTR at 24h.

Recommendation 1: Rebalance Marketing Acquisition Budgets

- **Observation:** Broad digital awareness campaigns perform worse than targeted outreach. The NovaEdge Awareness campaign incurred \$1.52M in spend while generating only \$34.3K in attributed pipeline. Conversely, Direct Mail and Partner Referrals delivered response rates of 53.0% and 34.3%.
- **Prescription:** Reallocate \$500,000 from underperforming broad email and awareness budgets directly into Account-Based Marketing (ABM), direct mail gifting, and co-marketing partner programs to drive higher-quality enterprise pipeline.

Recommendation 2: Institute an Enterprise Deal-Desk & Battle Card Library

- Observation: Sales cycle velocity averages 67 days, but enterprise accounts face significant friction. A review of the 184 lost deals shows that Pricing objections and Feature Gaps are the main reasons deals stalled.
- Prescription: Formalize an executive deal-desk workflow empowering sales leaders to approve structured multi-year pricing discounts and flexible payment terms. Also provide competitive battle cards to help account executives handle competitor feature comparisons.

Recommendation 3: Deploy an Enterprise Technical Escalation Pod

- Observation: Enterprise clients generated 757 support requests, with more than one out of four tickets expressing negative customer sentiment (25.5%). Critical system incidents in the Analytics Dashboard averaged over 102 hours to resolve—nearly double the organizational average of 58.2 hours.
- Prescription: Create a dedicated Tier-3 engineering escalation team focused on resolving enterprise outages quickly. Enforce a strict 24-hour MTTR SLA for critical dashboard and authentication issues to protect recurring contracts ahead of annual renewal conversations.